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The Editors-in-Chief
Advances in Structural Engineering
Sage

Dear Editors,

Submission of an original research article: “Blind spots and sensor placement for influence-area weigh-in-motion on continuous girders”

We submit the above manuscript for consideration as an Original Research Article in *Advances in Structural Engineering*. It falls within the journal’s focus on structural health monitoring for performance assessment of bridges.

Monitoring the loads that a bridge actually carries is often done with few channels, by integrating one response record and dividing by a calibration constant. On continuous girders this constant can vanish, because influence lines change sign over the piers. The manuscript gives structural engineers criteria for choosing, before installation, channels and positions that observe the load, and for judging whether their constant can be trusted when the pier stiffness is uncertain. The underlying identity and the generalized least-squares estimator are classical, and the manuscript says so; the contributions are the criteria built on them:

1. A closed-form catalog of blind loci for five channel types on prismatic two- and three-span girders. It shows that strain gauges that measure global bending (on a girder flange or web) are blind at every zero of the uniform-load moment and nearly blind in a band that can contain gauges set back from the piers. The strain criterion also holds for haunched girders, and the strain and cancellation indicators need no reference rigidity.

2. A placement rule from extreme-point optimal design: for position-independent noise, optimal sensors lie at the extremes of the uniform-load response. On one day of public moving-truck records, weighting gauges by their vehicle-free noise covariance raised the out-of-sample error above that of a plain gauge average.

3. An identifiability analysis under uncertain pier stiffness. For a symmetric girder with equal piers, the constant follows a linear-fractional (Möbius) law. On an in-service 55 + 90 + 55 m box girder, two static readings leave its side-span deflection constant

uncertain by a factor of 6.6 against 1.5 for the main-span constant from the static data alone, or 2.4 against 1.1 with girder rigidity bounded to 0.9–1.3 times the design value. At the design pier stiffness, a pier tiltmeter is the most informative of the additional measurements compared.

A laboratory test is consistent with the identity. The field records are used to check the assumptions behind the criteria; they cannot test weight discrimination, because their reference weights vary little, and the manuscript states this. The moving-load test of the in-service bridge is designed but not yet performed, and its predicted outcomes are given so that they can be checked.

The manuscript is just under 12 000 word-equivalents with 7 figures, 3 numbered tables and a table of symbols; derivations, full field tables and the error budget are in the Supplementary Material.

Data availability. The field strain records are from the public NOR-BWIM dataset (figshare, https://doi.org/10.7294/26880898.v1, CC0 license). The laboratory data, the processed data of the in-service bridge, the derived data behind the figures and tables, and all analysis scripts with their console output are provided as Supplementary Material and archived on Zenodo (https://doi.org/10.5281/zenodo.23535533).

The manuscript is original, has not been published, and is not under consideration by any other journal. All authors have approved the submission. Conflicts of interest, funding and the use of generative AI are declared in the manuscript.

Sincerely,

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